

Uncertainty quantification of shock wave equation of state (EOS): Basic theory and extensions

Condensed Matter Section (CMS) Seminar
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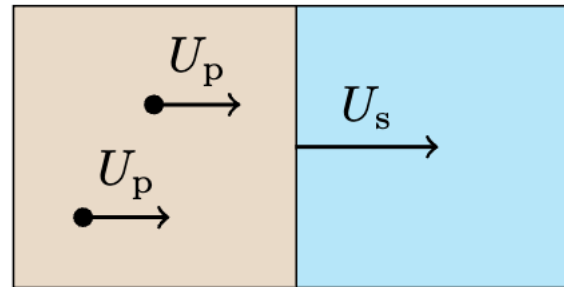
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Background on shock compression

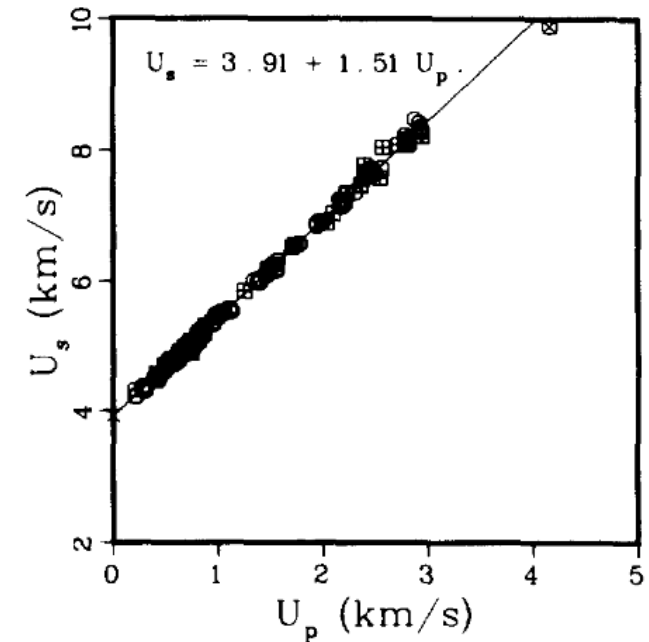


John Borg lab, <https://www.eng.mu.edu/shockphysics/capabilities/>

- Measure velocity of shock wave and material behind shock front



- U_p = particle velocity
- U_s = shock wave velocity



Copper data from Marsh (1980)

Want to quantify uncertainty for tasks like constructing EOS tables for hydrocode simulations.

Overview of presentation

- Quantifying uncertainty in linear model fits
- Propagating uncertainty to Hugoniot and isentropes
- Accounting for uncertainty in particle velocity

Details in Bernstein et al. (2026), J. of Applied Physics.

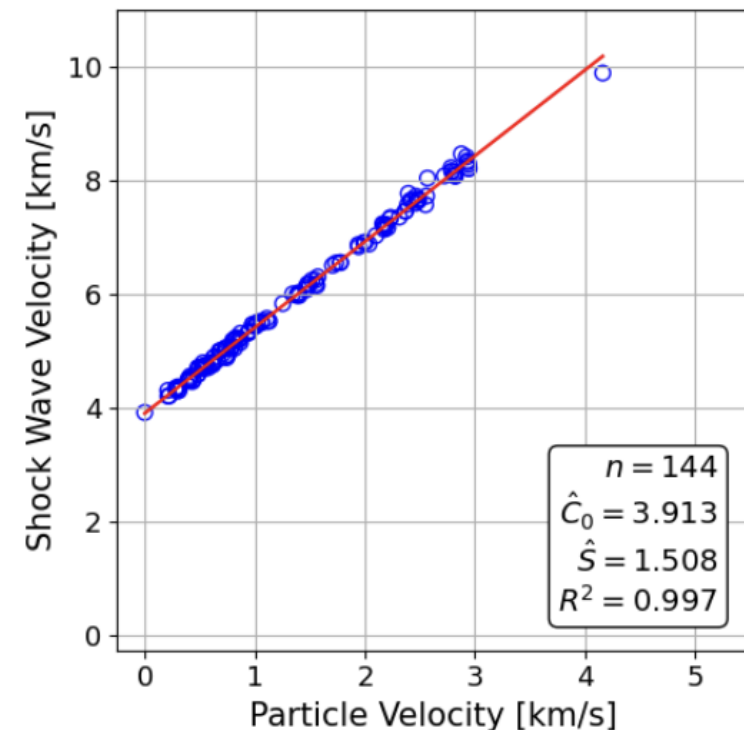
Experimental data (Marsh, 1980)

- Many materials exhibit linear relationship
- Modeled as

$$Y_i = C_0 + Sx_i + \epsilon_i$$

for $i = 1, \dots, n$, where Y_i is shock wave velocity and x_i is particle velocity.

- Least squares fits are common



What if we have prior information about parameters we want to account for?

Bayesian linear regression

- Rewrite model in standard notation as

$$Y = X\beta + \epsilon,$$

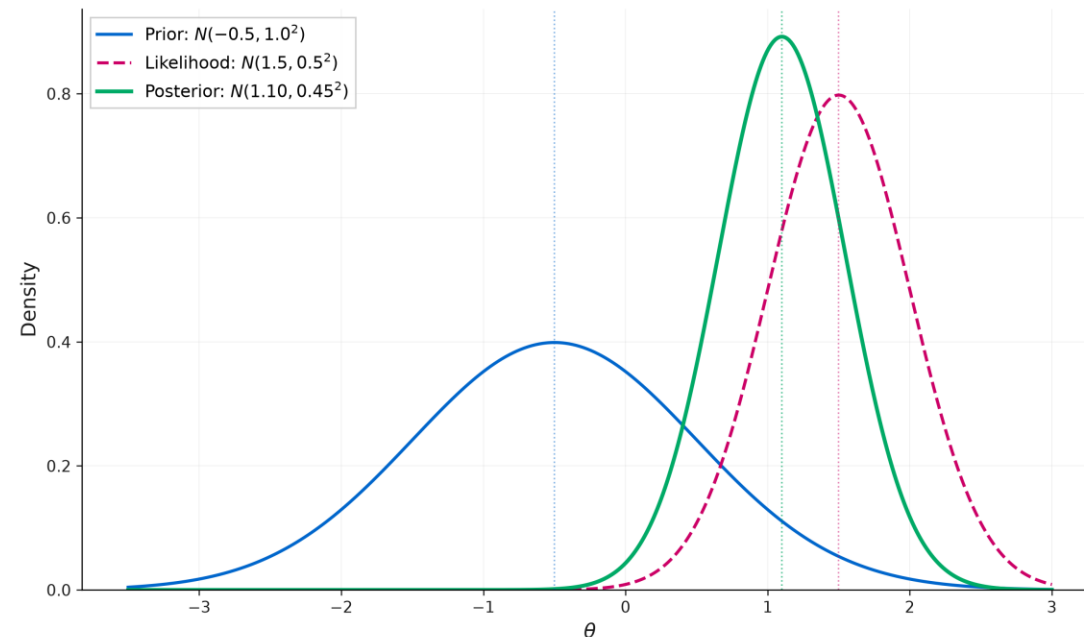
where $\beta = (C_0, S)'$ and $Y = (Y_1, \dots, Y_n)'$.

- Assume a non-informative prior distribution

$$p(\beta, \sigma^2) \propto \sigma^{-2}.$$

- Want posterior distribution

$$p(\beta, \sigma^2 | Y) \propto p(Y | \beta, \sigma^2) p(\beta, \sigma^2).$$



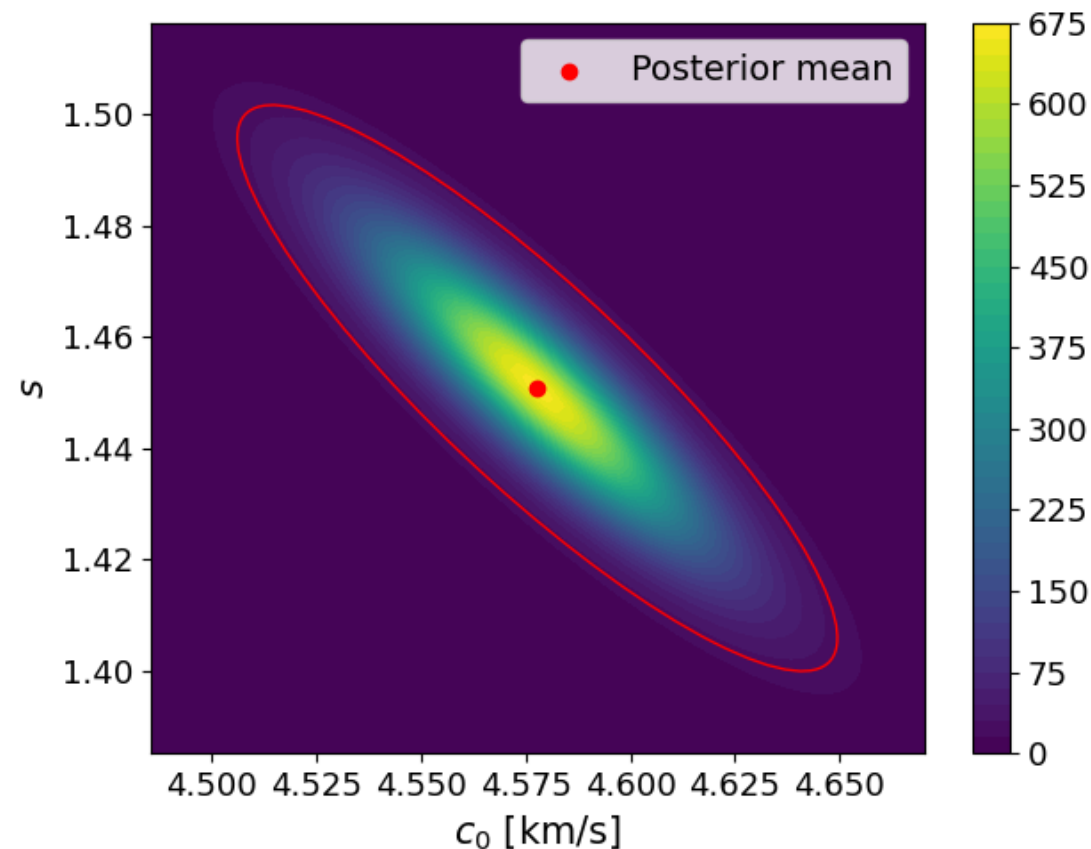
Joint posterior distribution is a normal-inverse-gamma distribution.

Analytical posterior distribution

- Marginal posterior distribution is in closed-form

$$\beta|Y \sim t_2(\hat{\beta}, \Sigma, \nu).$$

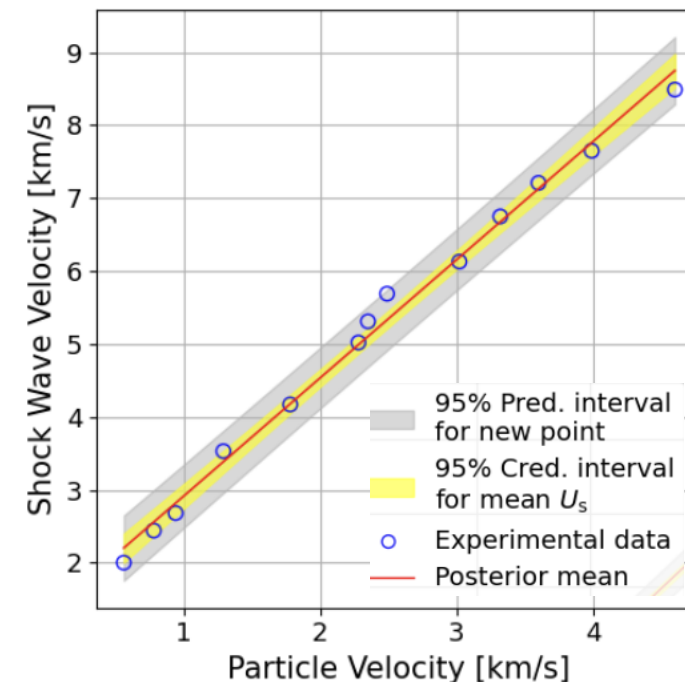
- This is a bivariate t-distribution
 - Centered at least squares estimate of β
 - Degrees of freedom is ν
 - Covariance matrix is $\frac{\nu}{\nu-2} \Sigma$
- Converges to normal distribution



Can evaluate and draw samples from distribution.

Propagating uncertainty through to shock wave velocity

- Two distributions of interest here
 1. Posterior distribution of mean shock wave velocity
 2. Prediction interval for new measurement
- Both univariate t-distributions



Argon data from Marsh (1980)

Uncertainty in mean decreases with n , variance converges to σ^2 .

Propagating uncertainty through Rankine-Hugoniot eqns.

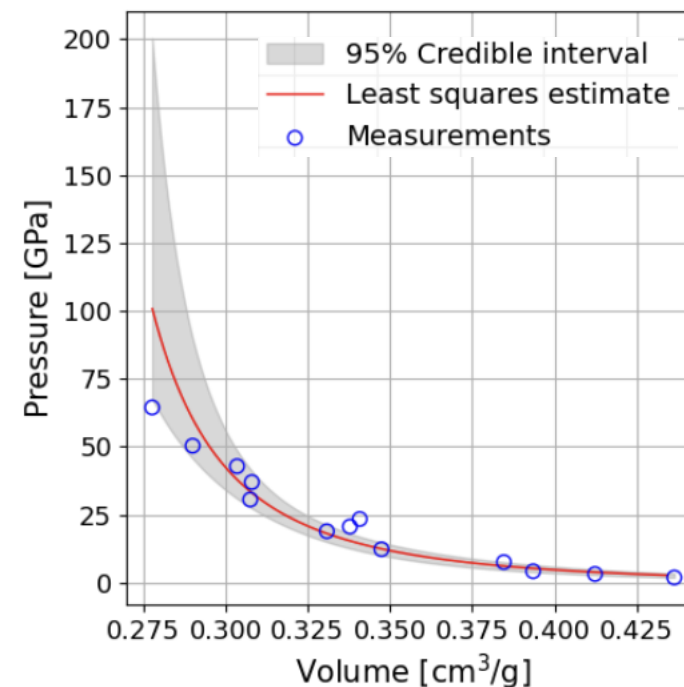
- Conservation of mass

$$\frac{V}{V_0} = \frac{U_s - U_p}{U_s}$$

- Conservation of momentum

$$P - P_0 = \rho_0 U_s U_p$$

- No analytical propagation since RH equations are non-linear
- Compute with Monte Carlo sampling instead

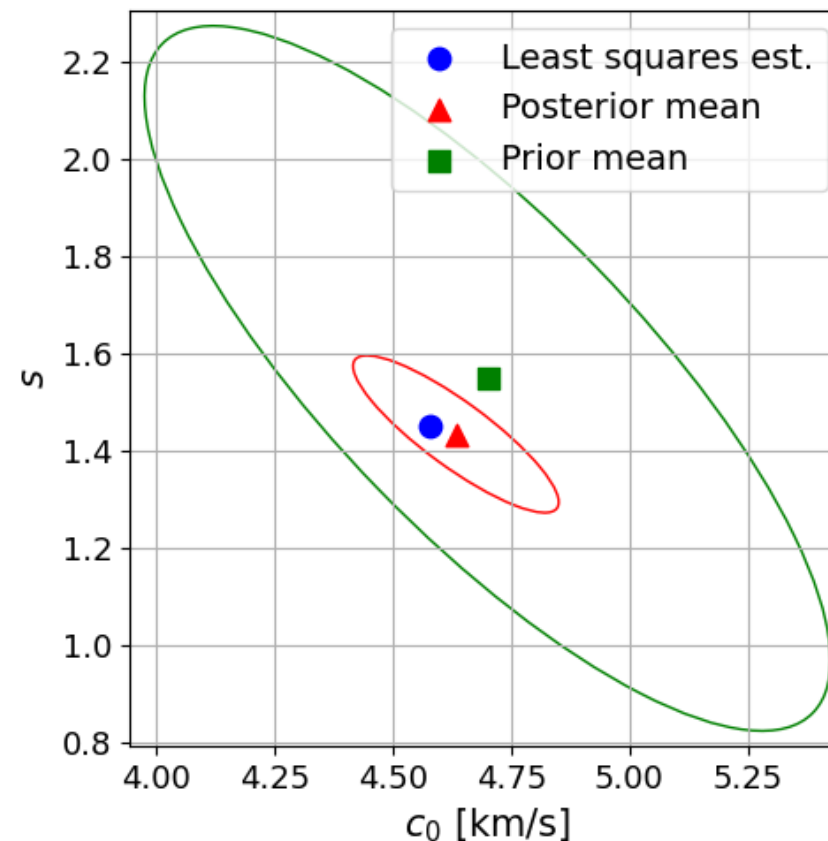


Argon data from Marsh (1980)

High-uncertainty at low volumes due to multiplicative equation for pressure.

Posterior with an informative prior distribution

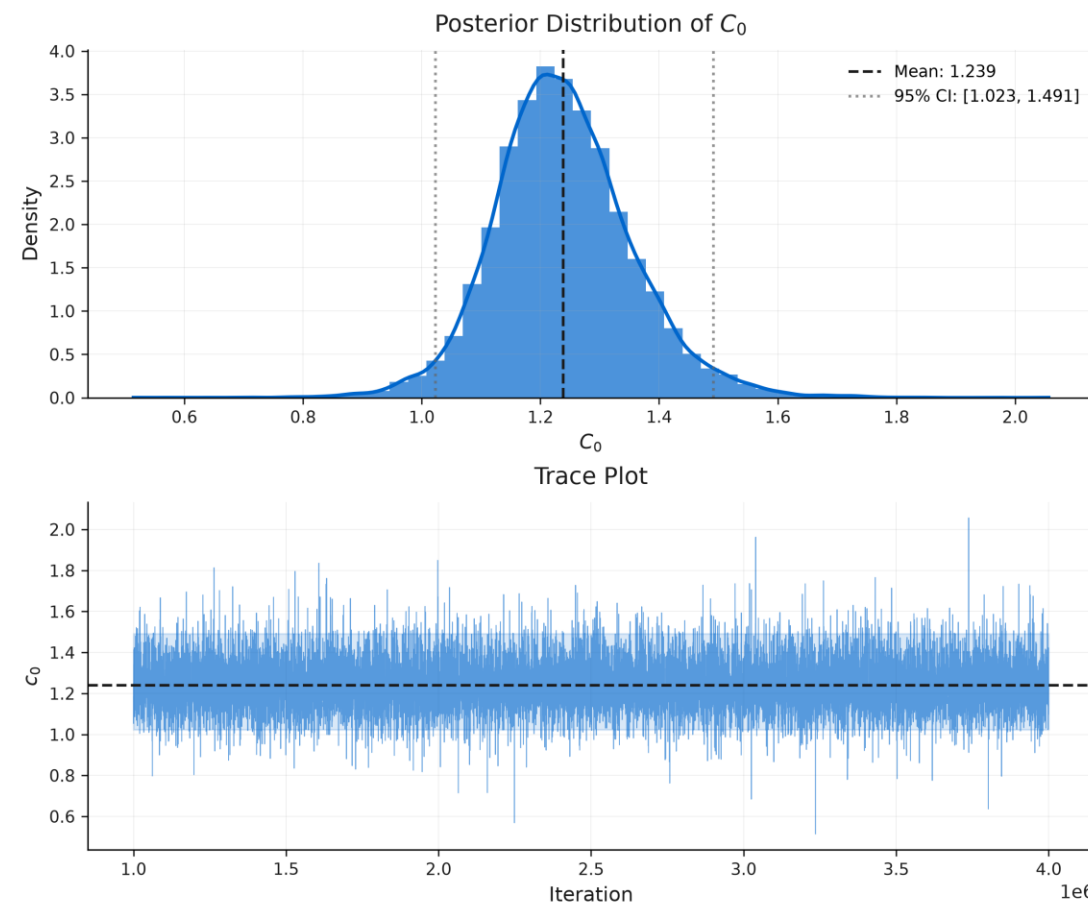
- Assume a normal prior distribution on $\beta = (C_0, S)'$
- Can show marginal posterior distribution of β is still a bivariate t-distribution
- Posterior mean is weighted average of prior mean and least squares estimate



See paper for all the details!

What about non-normal prior distributions on β ?

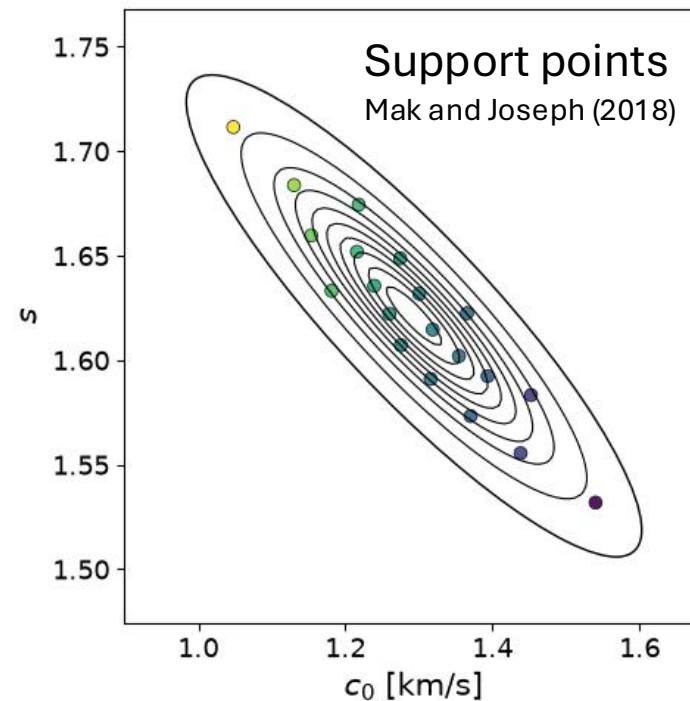
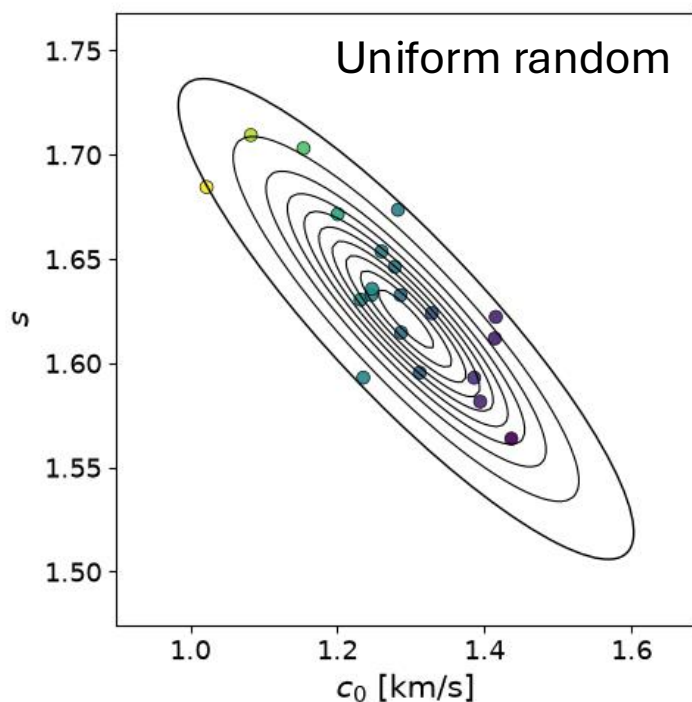
- Alternatively, can use Markov Chain Monte Carlo (MCMC) to sample posterior distribution
- An iterative, stochastic algorithm
- Not necessary since posterior can be sampled directly, but not wrong



Pros and cons to using MCMC for this problem.

Propagating uncertainty with limited samples

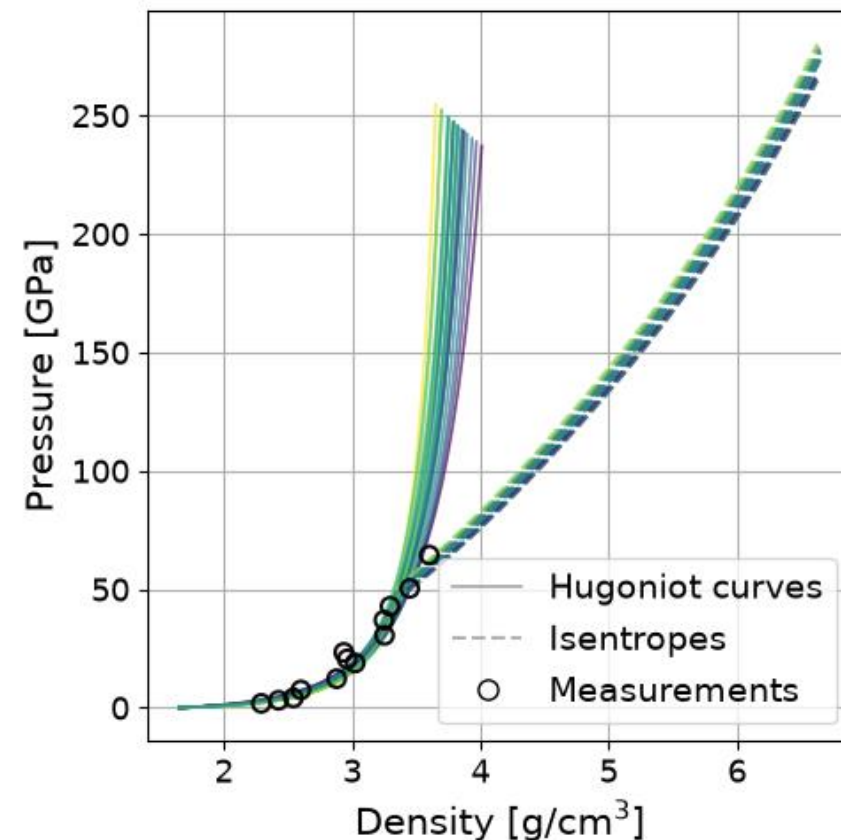
- How should we pick a small number of samples?



Support points tend to be more ‘spread out’ and have good convergence properties.

Propagating support points through LEOS code

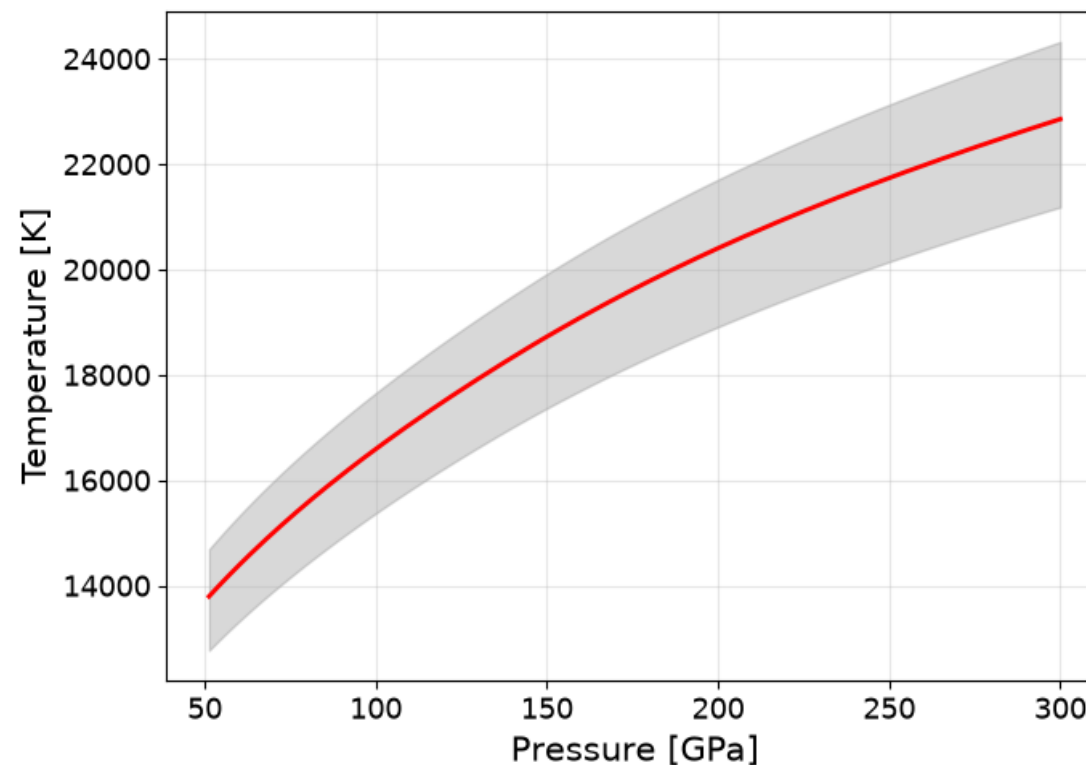
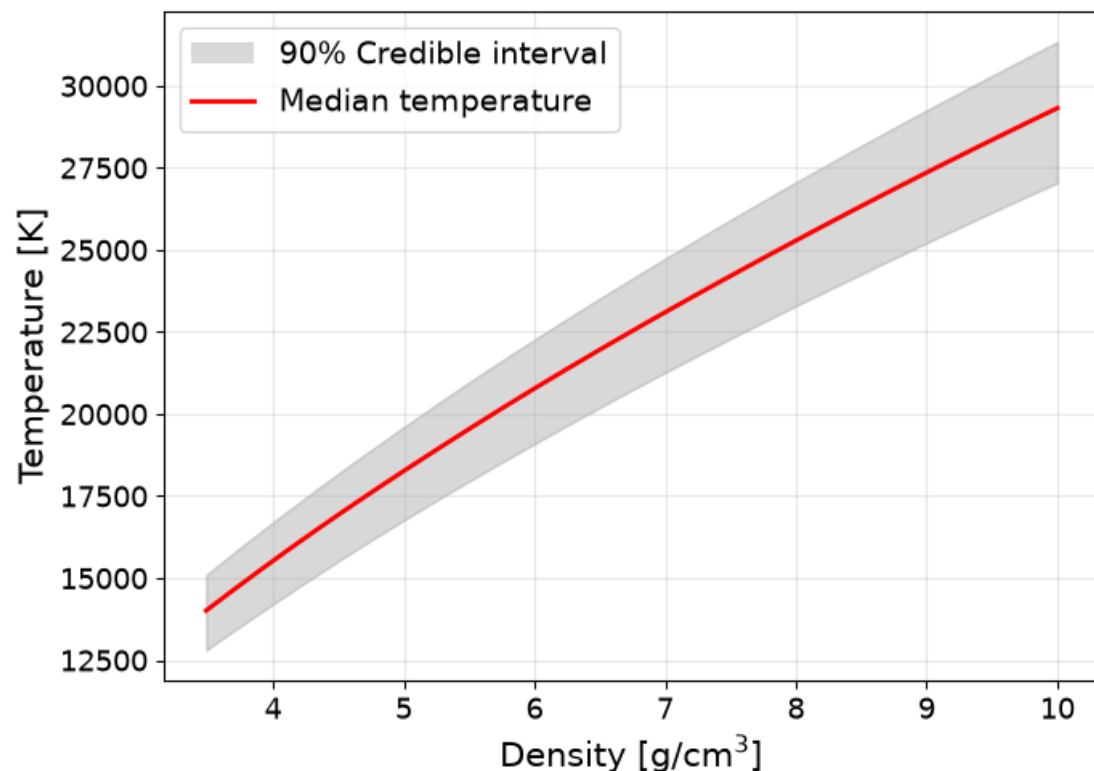
- 20 isentropic paths
- Each starts from a different Hugoniot curve at 50 GPa and goes to $\rho = 10 \text{ g/cm}^3$
- Together, they represent possible shock-ramp loading paths, assuming the material is first shocked to 50 GPa and then ramp-compressed.



Provides insight into material behavior during ramp compression.

Accessing temperature along the isentrope

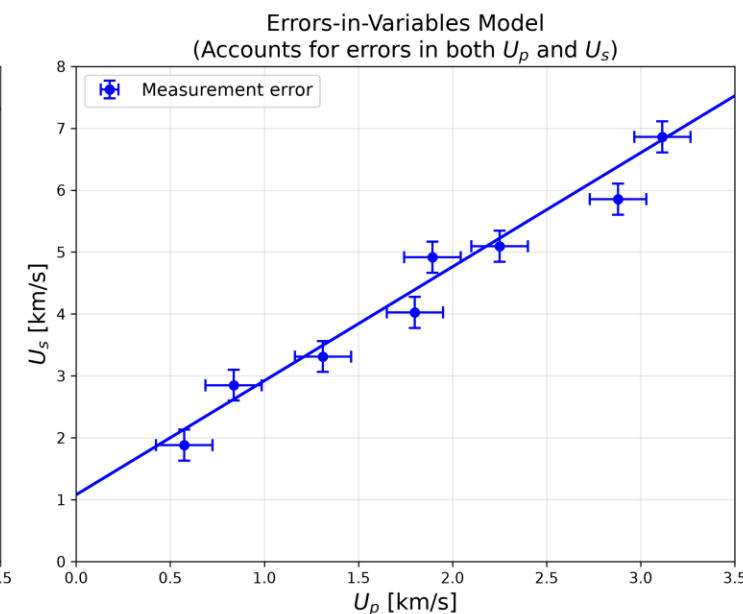
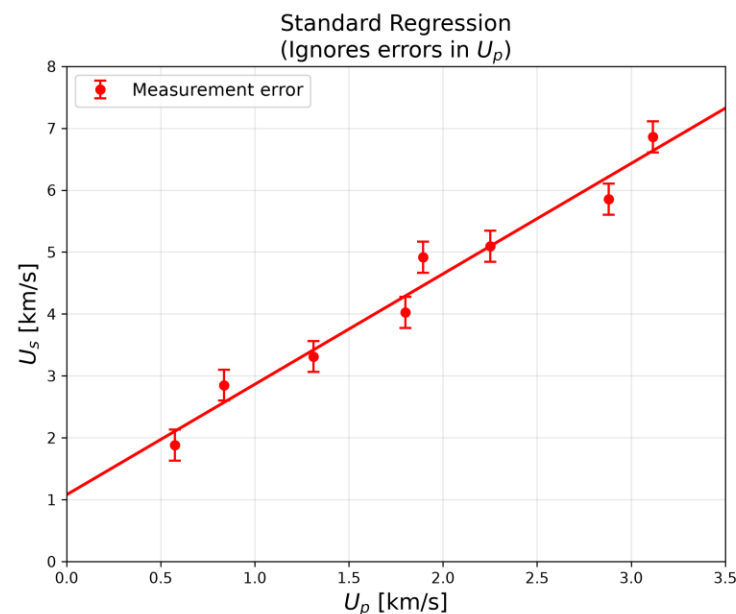
- Difficult to access temperature experimentally



Support points provide efficient means to assess temperature uncertainty.

Accounting for uncertainty in particle velocity

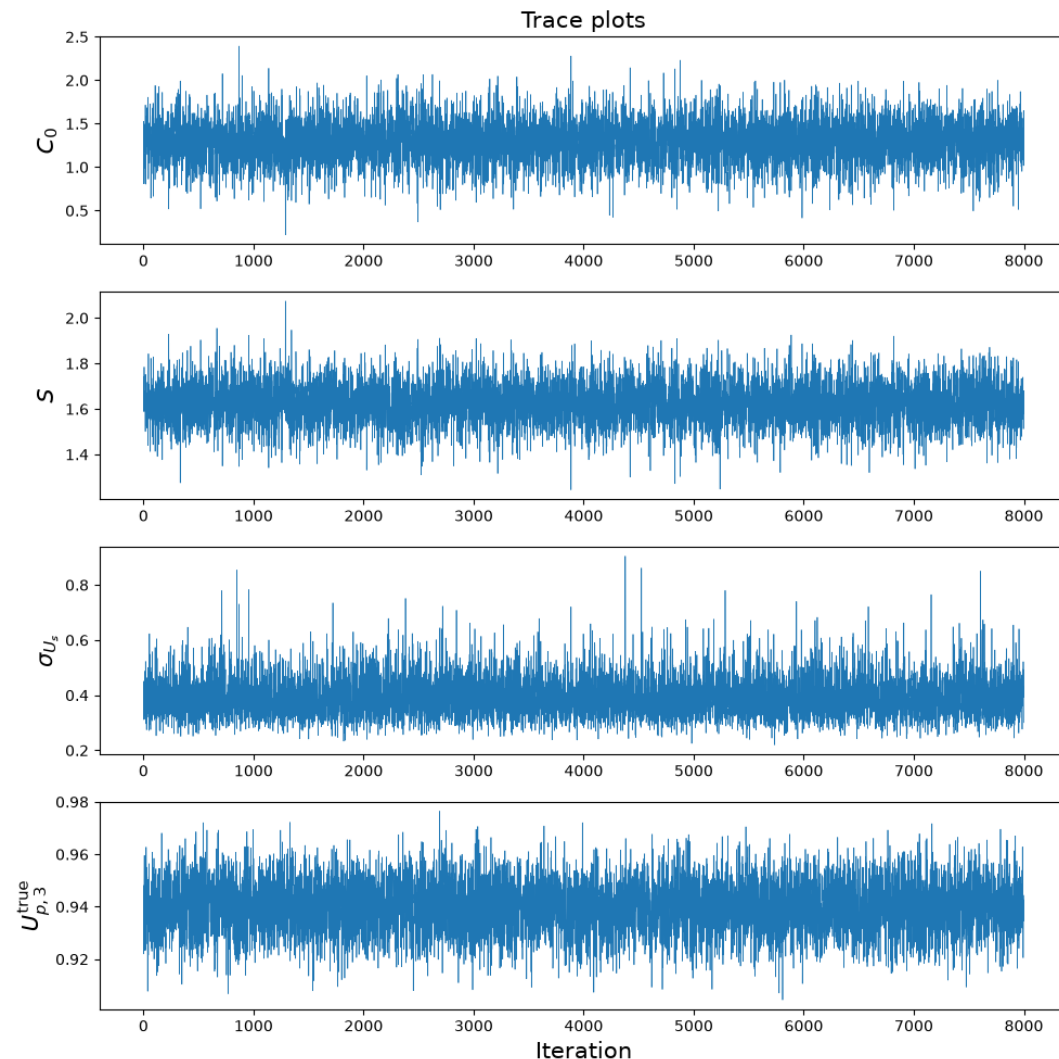
- Assume particle velocity is contaminated by Gaussian measurement error
- Called an error-in-variables (EIV) model
- Assume variance in U_p errors is known to avoid identifiability issue



Model accounts for more sources of uncertainty but is less trivial to fit.

Model inference

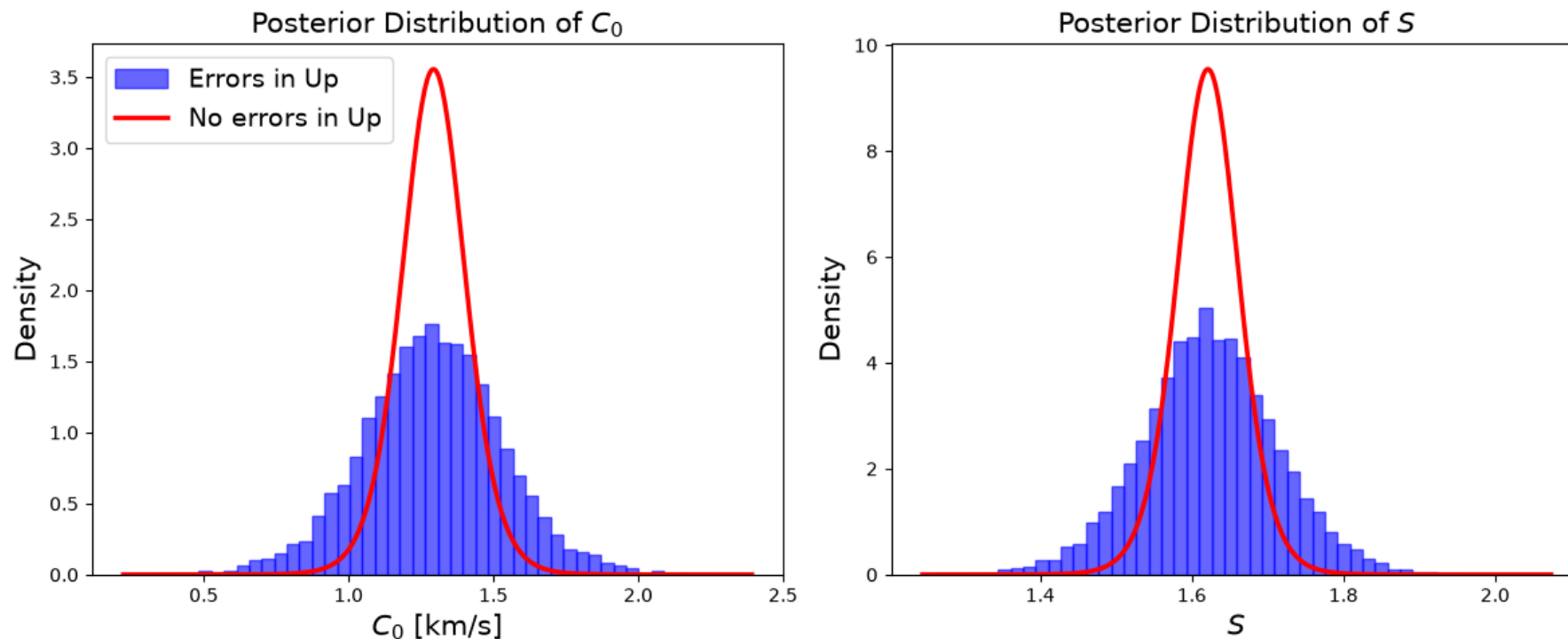
- Posterior distribution, $p(\beta, \sigma^2 | Y)$, is no longer available in closed-form
- Can be sampled using MCMC, specifically a Gibbs sampler
- Idea is to sample the posterior distribution of $\beta, \sigma_{U_s}^2, \sigma_{U_p}^2$, and the true particle velocities together
- Samples of a U_p $\longrightarrow$



Intuition: conditioning on U_p samples reduces model to previous case.

Comparing estimates for both models

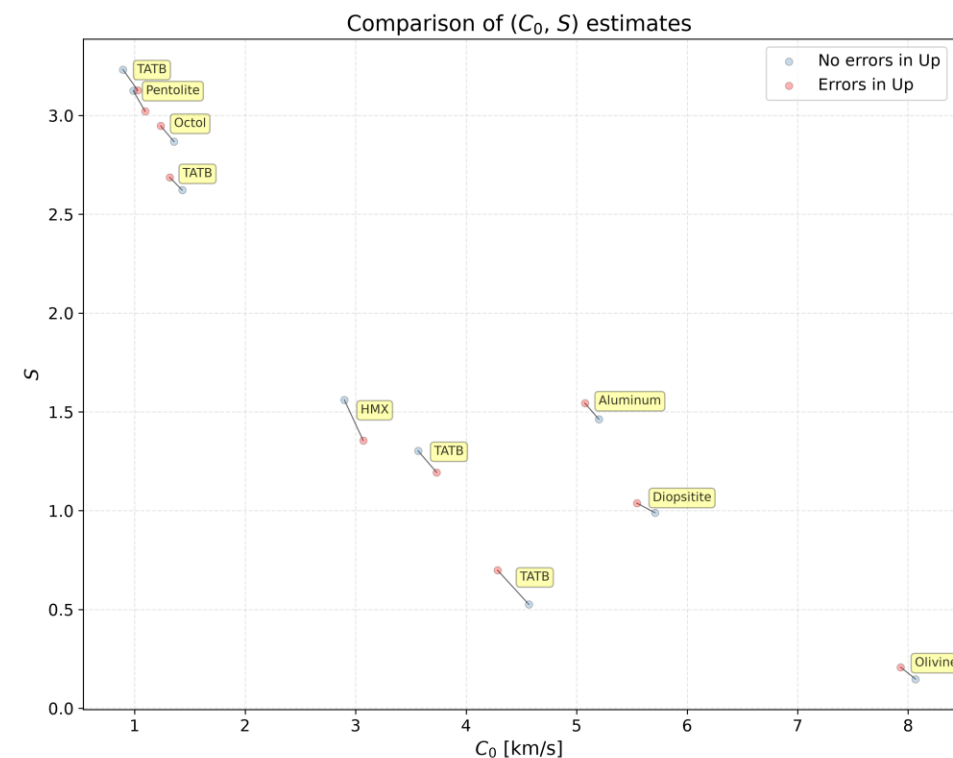
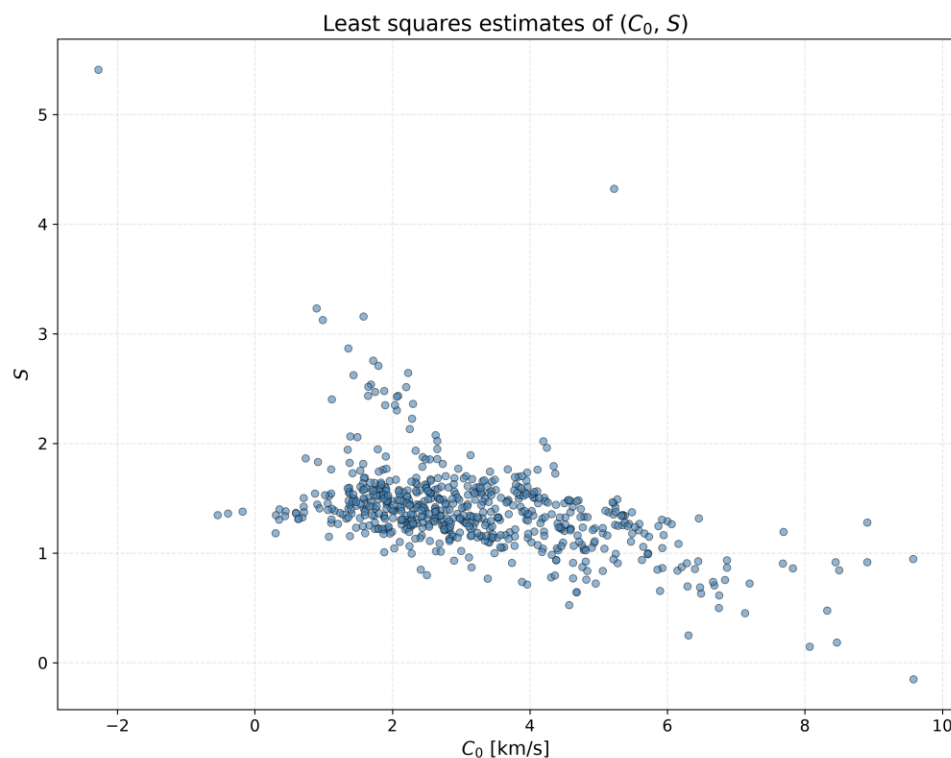
- Uncertainty in particle velocity increases uncertainty in C_0 and S parameters



Not seeing significant shrinking of S towards zero (attenuation bias).

Application to material shock wave database

- Repository of shock wave velocity data for ~800 materials



Attenuation is a common result, a decrease in S here.

Conclusions

- Bayesian inference is tractable for these linear models, though non-trivial to get details right
- Support points are smaller sample from posterior distribution that can be passed to expensive simulations
- Extensions not covered, but also doable:
 - Changes in linear slope
 - Non-linear models contaminated by measurement error
- GitHub repo: <https://github.com/llnl/balscd>

Thanks for listening!

References

- Bernstein, J., Myint, P. C., Lindquist, B. A., & Brown, J. L. (2026). A Tutorial on Bayesian analysis of linear shock compression data. *Journal of Applied Physics*, 140(3).
- Carroll, R. J., Ruppert, D., & Stefanski, L. A. (1995). *Measurement error in nonlinear models* (Vol. 105). CRC press.
- Mak, S., & Joseph, V. R. (2018). Support points. *The Annals of Statistics*, 46(6A), 2562–2592.
- Marsh, S. P. (1980). *LASL shock Hugoniot data* (Vol. 5). Univ of California Press.

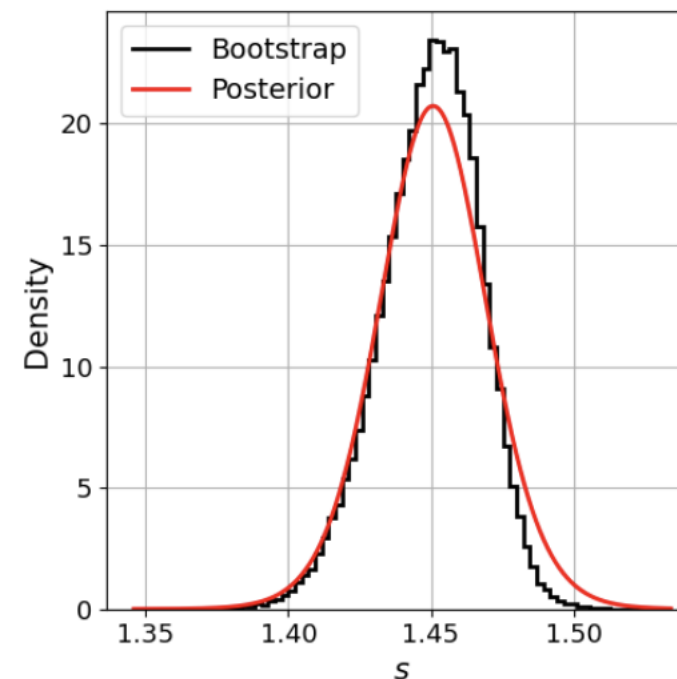
Thanks for listening!



Back-up slides

Comparison with bootstrapping

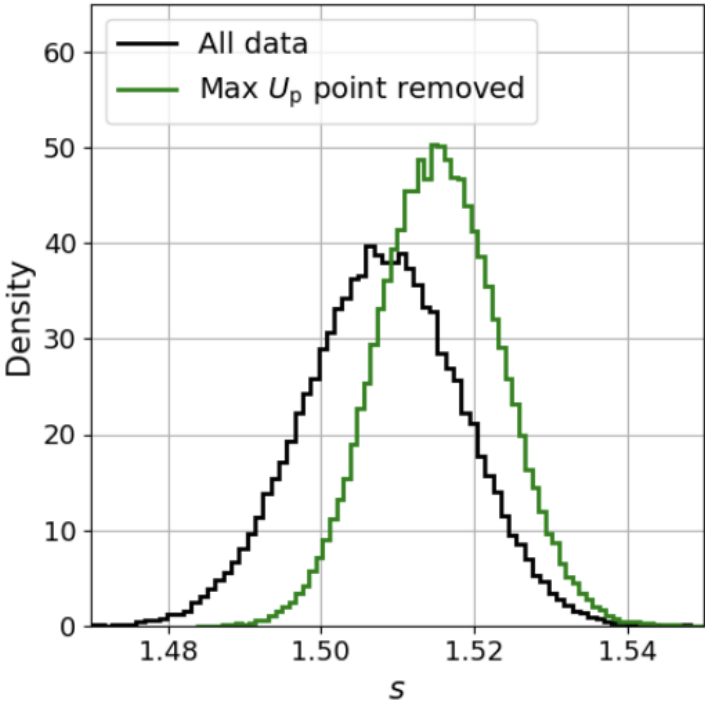
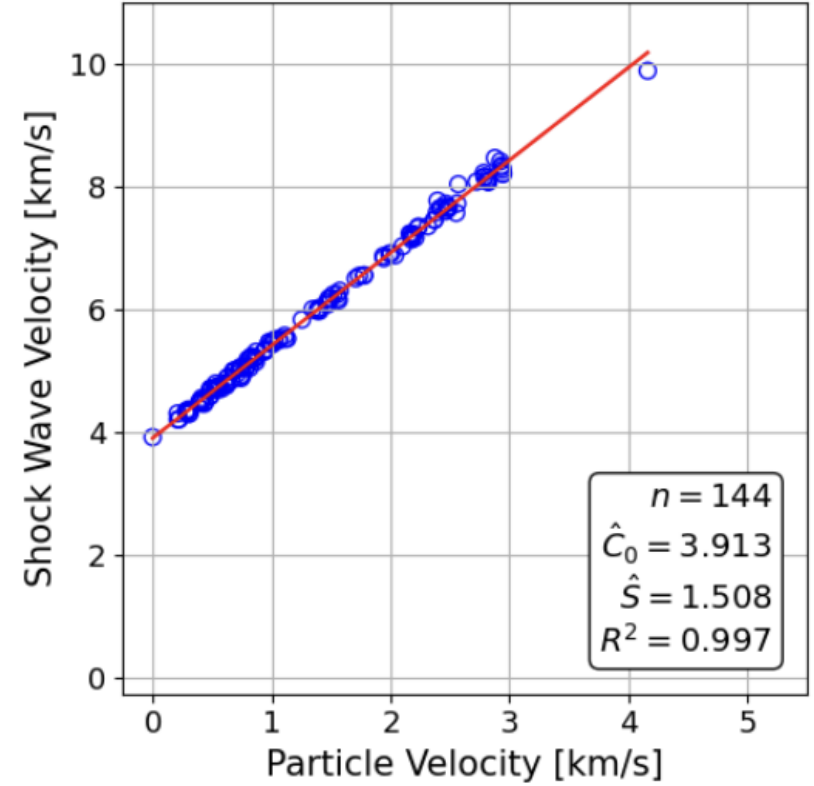
- Resampling-based approach
- Bootstrap distribution is not analytical or symmetric
- Good when don't want to make parametric assumptions or posterior not in closed-form
- Not in closed-form when prior or measurement errors not Gaussian



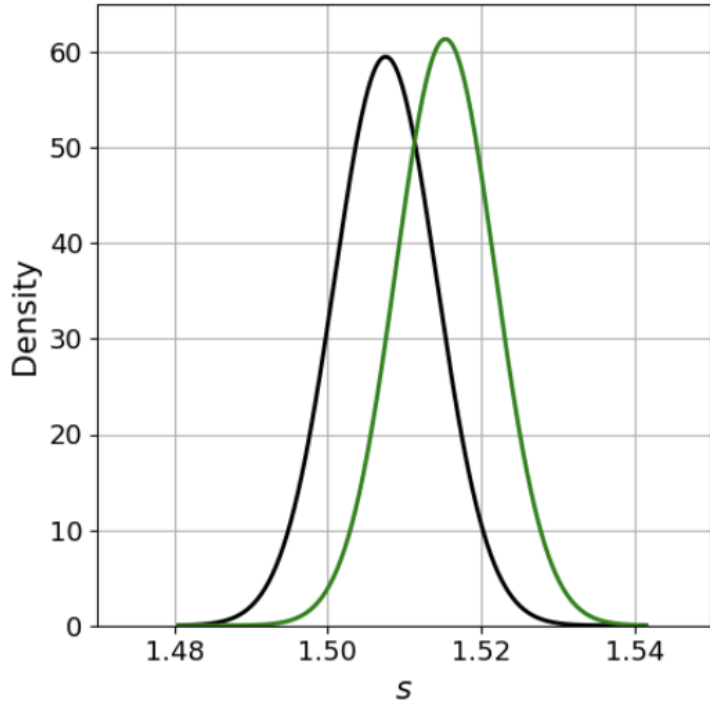
Nickel data [Marsh, 1980]

Key difference is in interpretation of bootstrap and posterior distributions.

Sensitivity analysis



(a) Bootstrap Distribution



(b) Posterior Distribution

Bootstrap distribution more sensitive to point with max particle velocity than posterior distribution.

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